

Generation Interconnection System Impact Study Report

For Queue Project Q24-2371

Boeing Renton 2 100 kV

June 20, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2371. The Interconnection Customer proposes a 50.0 MW battery energy storage facility interconnecting at the Boeing Renton 2 100 kV substation in the Indiana area. The request was received on January 03, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 434 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,068,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: S Whitehall - Roseau County 100.0 kV Line
Contingency: Dekalb - Richland 500.0 kV
Loading: 133.8% (Rating: 172 MVA)
DFAX: 14.6%
Pre-project loading: 129.6%
2. Facility: State St - W Olympia 100.0 kV #1
Contingency: Boeing Renton 2 - S Whitehall 100.0 kV
Loading: 112.6% (Rating: 167 MVA)
DFAX: 5.2%
Pre-project loading: 111.0%
3. Facility: Stae Street Substation - W Olympia 100.0 kV Ckt 2
Contingency: Boeing Renton 2 - South Whitehall 100.0 kV
Loading: 112.6% (Rating: 167 MVA)
DFAX: 5.2%
Pre-project loading: 111.0%
4. Facility: W Olympia - State St 100.0 kV
Contingency: Boeing Renton2 - South Whitehall 100.0 kV
Loading: 112.6% (Rating: 167 MVA)
DFAX: 5.2%
Pre-project loading: 111.0%
5. Facility: Peavine - South Whitehall Substation 100.0 kV
Contingency: Peavine Substation - S Whitehall 100.0 kV Line 2
Loading: 107.3% (Rating: 149 MVA)
DFAX: 44.3%
Pre-project loading: 92.4%
6. Facility: S Whitehall - Peavine 100.0 kV Ckt 2
Contingency: South Whitehall - Pavine 100.0 kV Circuit 1
Loading: 107.3% (Rating: 149 MVA)

DFAX: 44.3%
Pre-project loading: 92.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S Whitehall - Roseau County 100.0 kV Line	Rebuild line with higher-capacity conductor	\$12,594,000	53% (\$6,653,000)
State St - W Olympia 100.0 kV #1	Rebuild line with higher-capacity conductor	\$1,317,000	21% (\$272,000)
State Street - W Olympia 100.0 kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,326,000	21% (\$274,000)
W Olympia - State St 100.0 kV	Rebuild line with higher-capacity conductor	\$837,000	21% (\$173,000)
S Whitehall - Peavine 100.0 kV	Reconductor line	\$3,423,000	83% (\$2,851,000)
S Whitehall - Peavine 100.0 kV Ckt 2	Reconductor line	\$3,416,000	83% (\$2,845,000)
	Total allocated to project		\$13,068,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.